function also containing two poles. The problem is, the substitution leaves the transfer function in a very unfriendly form:

$$H(z) = \frac{1}{\left(\frac{2(1-z^{-1})}{T(1+z^{-1})} - (\sigma + j\omega)\right)\left(\frac{2(1-z^{-1})}{T(1+z^{-1})} - (\sigma - j\omega)\right)}$$

Working through the long and tedious algebra, this expression can be placed in the standard form of Eq. 33-3, and the recursion coefficients identified as:

$$\begin{aligned} a_0 &= T^2/D \\ a_1 &= 2T^2/D \\ a_2 &= T^2/D \\ b_1 &= (8 - 2MT^2)/D \\ b_2 &= (-4 - 4\sigma T - MT^2)/D \end{aligned}$$

where:

$$\begin{aligned} M &= \sigma^2 + \omega^2 \\ T &= 2\tan(1/2) \\ D &= 4 - 4\sigma T + MT^2 \end{aligned}$$

EQUATION 33-11
Bilinear transform for two poles. The pole-pair is located at $\sigma \pm \omega$ in the s-plane, and a_0, a_1, a_2, b_1, b_2 are the recursion coefficients for the discrete system.

The variables *M*, *T*, and *D* have no physical meaning; they are simply used to make the equations shorter.

Lines 1200-1290 use these equations to convert the location of the s-domain pole-pair, held in the variables, RP and IP, directly into the recursive coefficients, held in the variables, X0, X1, X2, Y1, Y2. In other words, we have calculated an intermediate result: the recursion coefficients for one stage of a *low-pass* filter with a cutoff frequency of *one*.

Low-pass to Low-pass Frequency Change

Changing the frequency of the recursive filter is also accomplished with a conformal mapping technique. Suppose we know the transfer function of a recursive low-pass filter with a unity cutoff frequency. The transfer function of a similar low-pass filter with a new cutoff frequency, *W*, is obtained by using a **low-pass to low-pass transform**. This is also carried